

Week 11 – Advanced Machine Learning Models

Customer Segmentation & Prediction

1. Project Overview

This project focuses on customer segmentation and churn prediction using machine learning

The main objectives are:

- Segment customers using clustering algorithms.
- Build prediction models for different customer segments.
- Evaluate model performance.
- Identify important features.
- Provide business recommendations.

2. Dataset

Dataset: `customer\_churn.csv`

The dataset contains 500 customer records with information such as:

- CustomerID
- Tenure
- MonthlyCharges
- TotalCharges
- Contract
- PaymentMethod
- PaperlessBilling
- SeniorCitizen
- Churn

3. Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

4. Data Preprocessing

The following steps were performed:

- Checked missing values and duplicates.
- Removed/ignored CustomerID for modelling.
- Encoded categorical variables.
- Scaled numerical features where required.
- Prepared data for clustering and prediction.

5. Customer Segmentation

Multiple clustering techniques were used:

- K-Means
- Hierarchical Clustering
- DBSCAN

The Elbow Method and Silhouette Score were used to select an appropriate clustering solution.

6. Customer Profiles

After clustering, customers were grouped into different segments.

Each segment was analyzed based on:

- Number of customers
- Tenure
- Monthly charges
- Total charges
- Contract type
- Churn rate

Segment profiles were saved in the `results/` folder.

7. Churn Prediction

Separate prediction models were created for different customer segments.

Random Forest was used as the main classification algorithm.

The models were trained and tested using a train-test split.

8. Hyperparameter Tuning

Random Forest parameters were optimized using hyperparameter tuning.

The best-performing parameters were selected based on model evaluation results.

9. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC where applicable
- Confusion Matrix

The evaluation results are stored in the `results/` folder.

10. Feature Importance

Feature importance was calculated using the Random Forest models to identify the most useful features for churn prediction.

11. Business Recommendations

Based on customer segments and churn patterns:

- High-risk customers should receive targeted retention offers.
- Customer engagement should be improved for vulnerable segments.
- Long-term contracts and loyalty benefits can be promoted where appropriate.
- Segment-specific strategies can be used instead of a single strategy for all customers.

12. Visualizations

The project includes visualizations for:

- Elbow Method
- Silhouette Scores
- Customer Clusters
- Segment Profiles
- Feature Importance
- Model Performance

All visualizations are stored in the `visualizations/` folder.

```
File Edit Selection View Go Run ... Week1_Task
customer_segmentation.py dataset_summary.txt
Task 11 Advanced_ML_Models > customer_segmentation.py > train_segment_models
393 def train_segment_models(df: pd.DataFrame, segment_mapping: dict[int, str]) -> tuple[pd.DataFrame, pd.DataFrame]:
400     continue
401
Problems Debug Console Ports Spell Checker Code Reference Log
Terminal
sns.barplot(data=importance, x="importance", y="feature", palette="viridis")
D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\customer_segmentation.py:432: FutureWarning:
Passing 'palette' without assigning 'hue' is deprecated and will be removed in v0.14.0. Assign the 'y' variable to 'hue' and set 'legend=False' for the same effect.
sns.barplot(data=importance, x="importance", y="feature", palette="viridis")
=== MODEL EVALUATION RESULTS ===
segment      segment_name  accuracy  precision  recall  f1_score  roc_auc
0  Higher-Value Premium Base  1.000000    1.0 1.000000    1.0 1.000000 {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 2, "model_min_samples_split": 2, "model_n_estimators": 200} [{"num_Tenure", "num_TotalCharges", "num_MonthlyCharges", "cat_Contract_Two_Year", "num_SeniorCitizen"}]
1  Established Customer Base 1  0.944444    0.0 0.000000    0.0 1.000000 {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 1, "model_min_samples_split": 2, "model_n_estimators": 100} [{"num_Tenure", "num_TotalCharges", "num_MonthlyCharges", "num_SeniorCitizen", "cat_Contract_One_Year"}]
2  Highest Churn Priority      1.000000    1.0 1.000000    1.0 1.000000 {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 1, "model_min_samples_split": 2, "model_n_estimators": 100} [{"num_Tenure", "num_MonthlyCharges", "num_TotalCharges", "num_SeniorCitizen", "cat_PaymentMethod_Electronic_Check"}]
3  Lower-Charge Opportunity   0.941176    1.0 0.666667    0.8 1.000000 {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 1, "model_min_samples_split": 2, "model_n_estimators": 100} [{"num_Tenure", "num_MonthlyCharges", "num_TotalCharges", "cat_Contract_Month-To-Month", "cat_Contract_One_Year"}]
4  Lowest Churn Stability     0.944444    0.0 0.000000    0.0 NaN {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 2, "model_min_samples_split": 5, "model_n_estimators": 100} [{"num_Tenure", "num_TotalCharges", "num_MonthlyCharges", "num_SeniorCitizen", "cat_PaymentMethod_Credit_Card"}]
5  Established Customer Base 2  0.933333    0.0 0.000000    0.0 0.785714 {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 1, "model_min_samples_split": 2, "model_n_estimators": 100} [{"num_Tenure", "num_MonthlyCharges", "num_TotalCharges", "cat_PaymentMethod_Bank_Transfer", "num_SeniorCitizen"}]
=== FINAL SUMMARY ===
Optimal K: 6
Hierarchical silhouette: 0.1664
Cluster assignments saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\cluster_assignments.csv
Segment profiles saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\segment_profiles.csv
Model evaluation saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\model_evaluation_results.csv
Prediction results saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\prediction_results.csv
Markdown report saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\report\segment_profiles.md
D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models>

File Edit Selection View Go Run ... Week1_Task
customer_segmentation.py dataset_summary.txt
Task 11 Advanced_ML_Models > customer_segmentation.py > train_segment_models
393 def train_segment_models(df: pd.DataFrame, segment_mapping: dict[int, str]) -> tuple[pd.DataFrame, pd.DataFrame]:
400     continue
406     rows.append({
407         "segment": segment_id,
408     })
Problems Debug Console Ports Spell Checker Code Reference Log
Terminal
e1_min_samples_leaf": 2, "model_min_samples_split": 5, "model_n_estimators": 100} [{"num_Tenure", "num_TotalCharges", "num_MonthlyCharges", "num_SeniorCitizen", "cat_PaymentMethod_Credit Card"}]
5 Established Customer Base 2  0.933333    0.0 0.000000    0.0 0.785714 {"model_max_depth": null, "model_max_features": "sqrt", "model_min_samples_leaf": 1, "model_min_samples_split": 2, "model_n_estimators": 100} [{"num_Tenure", "num_MonthlyCharges", "num_TotalCharges", "cat_PaymentMethod_Bank Transfer", "num_SeniorCitizen"}]
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Segment profiles saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\segment_profiles.csv
Model evaluation saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\model_evaluation_results.csv
Prediction results saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\results\prediction_results.csv
Markdown report saved to: D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models\report\segment_profiles.md
D:\Developer_Arena\Week1_Task\Task_11_Advanced_ML_Models>
```

13. Project Structure

Task_11_Advanced_ML_Models/

```
|
|
|— customer_churn.csv
|— customer_segmentation.py
|— requirements.txt
|
|— results/
|— visualizations/
|— screenshots/
|— report/
    |— segment_profiles.md
```

14. Setup & Run

Install dependencies:

```
python -m pip install -r requirements.txt
```

Run the project:

```
python customer_segmentation.py
```

15. Conclusion

This project demonstrates an end-to-end machine learning workflow including data preprocessing, customer segmentation, churn prediction, hyperparameter tuning, model evaluation, feature importance analysis, and business recommendations.